

EXPERIMENT 2

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Question A:

(A):

You are given with the Employee schema; the task is to find the highest score for each respective department.

For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28. Same with the other departments i.e., D2, D3 and D4.

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

Solution:

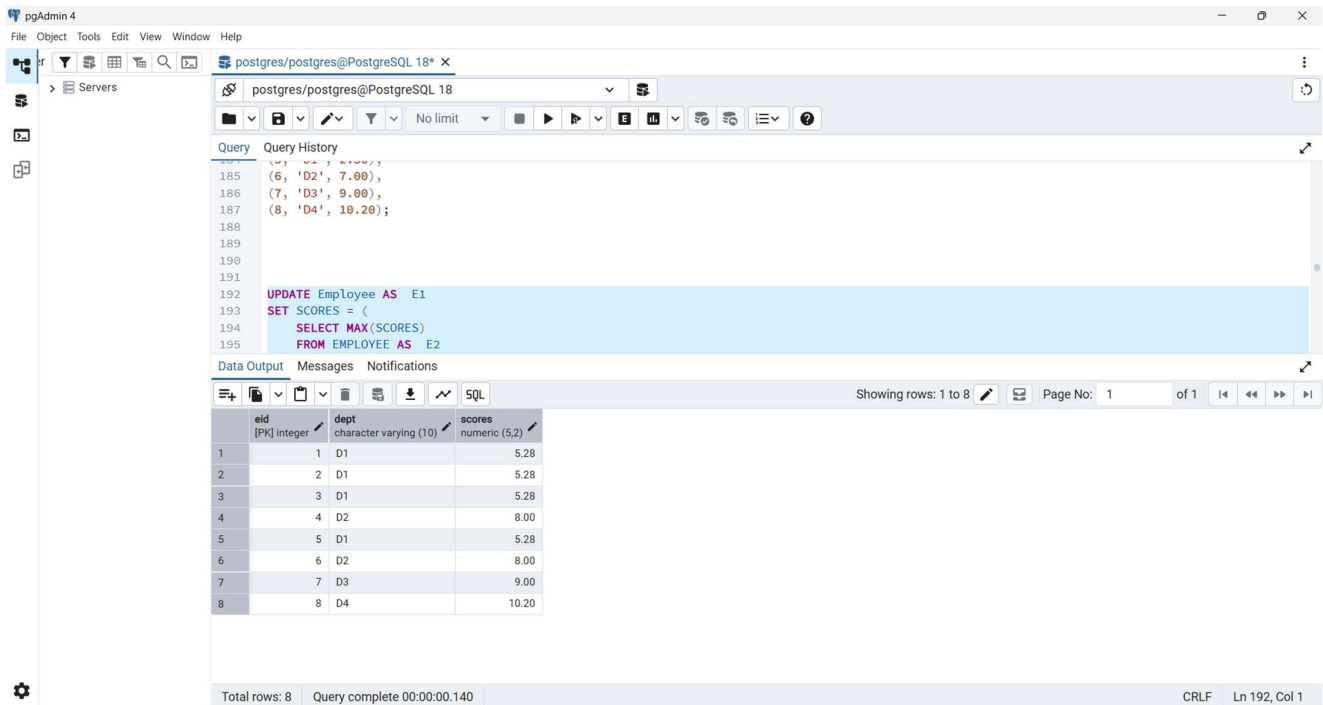
1. Query:

```
UPDATE Employee AS E1
SET SCORES = (
    SELECT MAX(SCORES)
    FROM EMPLOYEE AS E2
    WHERE DEPT = E1.DEPT
);

SELECT * FROM EMPLOYEE ;
```

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2. Output



The screenshot shows the pgAdmin 4 interface. The query editor displays the following SQL code:

```

185 (6, 'D2', 7.00);
186 (7, 'D3', 9.00);
187 (8, 'D4', 10.20);
188
189
190
191
192 UPDATE Employee AS E1
193 SET SCORES = (
194     SELECT MAX(SCORES)
195     FROM EMPLOYEE AS E2
  
```

The Data Output tab shows the following table:

eid	dept	scores
1	D1	5.28
2	D1	5.28
3	D1	5.28
4	D2	8.00
5	D1	5.28
6	D2	8.00
7	D3	9.00
8	D4	10.20

At the bottom, it indicates: Total rows: 8, Query complete 00:00:00.140, CRLF, Ln 192, Col 1.

Question B:

(C): SALES ANALYSIS PART 1

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

Query:

```

SELECT SELLER_ID
FROM ORDERS
GROUP BY SELLER_ID
HAV(SELECT
    MAX(MAX_AMT)
    FROM
    (SELECT SELLER_ID , SUM(AMOUNT) AS MAX_AMT
    FROM ORDERS
    GROUP BY SELLER_ID ))
  
```

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Output:

Data Output	Messages	Notifications									
+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL <table> <tr> <th></th><th>seller_id</th><th></th></tr> <tr> <td></td><td>integer</td><td>🔒</td></tr> <tr> <td>1</td><td>3</td><td></td></tr> </table>		seller_id			integer	🔒	1	3			
	seller_id										
	integer	🔒									
1	3										

Question C:

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

Query :

```

SELECT D.NAME AS DEPARTMENT,
       E.NAME AS EMPLOYEE ,
       E.SALARY AS SALARY
FROM EMPLOYEE AS E
JOIN DEPARTMENT AS D
ON E.DEPARTMENTID = D.ID
WHERE (
    SELECT COUNT(DISTINCT E2.SALARY)
    FROM Employee AS E2
    WHERE E2.departmentId = E.departmentId
    AND
    E2.SALARY > E.SALARY
) < 3
ORDER BY D.NAME ;

```

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Output:

Data Output Messages Notifications

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Joe	85000
2	IT	Max	90000
3	IT	Randy	85000
4	IT	Will	70000
5	Sales	Henry	80000
6	Sales	Sam	60000